

ECON4200

2026-04-24

Setting up

load library

earning code mapping

```
earnings_map <- tibble(  
  code = str_pad(1:40, width = 2, pad = "0"),  
  midpoint_hkd = c(1000,2500,3500,4500,5500,6500,7500,8500,9500,  
                   10500,11500,12500,13500,14500,15500,16500,17500,  
                   18500,19500,21000,23000,25000,27000,29000,31000,  
                   33000,35000,37000,39000,41000,43000,45000,47000,  
                   49000,55000,65000,75000,85000,95000,120000)  
)  
  
other_earnings_map <- tibble(  
  code = str_pad(1:11, width = 2, pad = "0"),  
  midpoint_hkd = c(500,1500,2500,3500,4500,5500,6500,7500,8500,9500,12000)  
)
```

Load all files + calculate est hourly wage

survey does not give hourly wage directly Monthly earning is in code e.g., code “01” = < HK\$2,000, code “02” = HK\$2,000–2,999, etc. Hours worked in the last 7 days (or in main employment)

convert the monthly banded earnings into an approximate hourly rate

using midpoint of ranges divide hours work will give hourly wage

hours_main_employment = “Hours worked in main employment (for employed persons)” → This is hours worked in the main job only during the past 7 days.

hours_main_employment * 4.345 is monthly working hour

hours_7days = “Hours worked during the seven days before enumeration” → This is total hours across all jobs in the past 7 days.

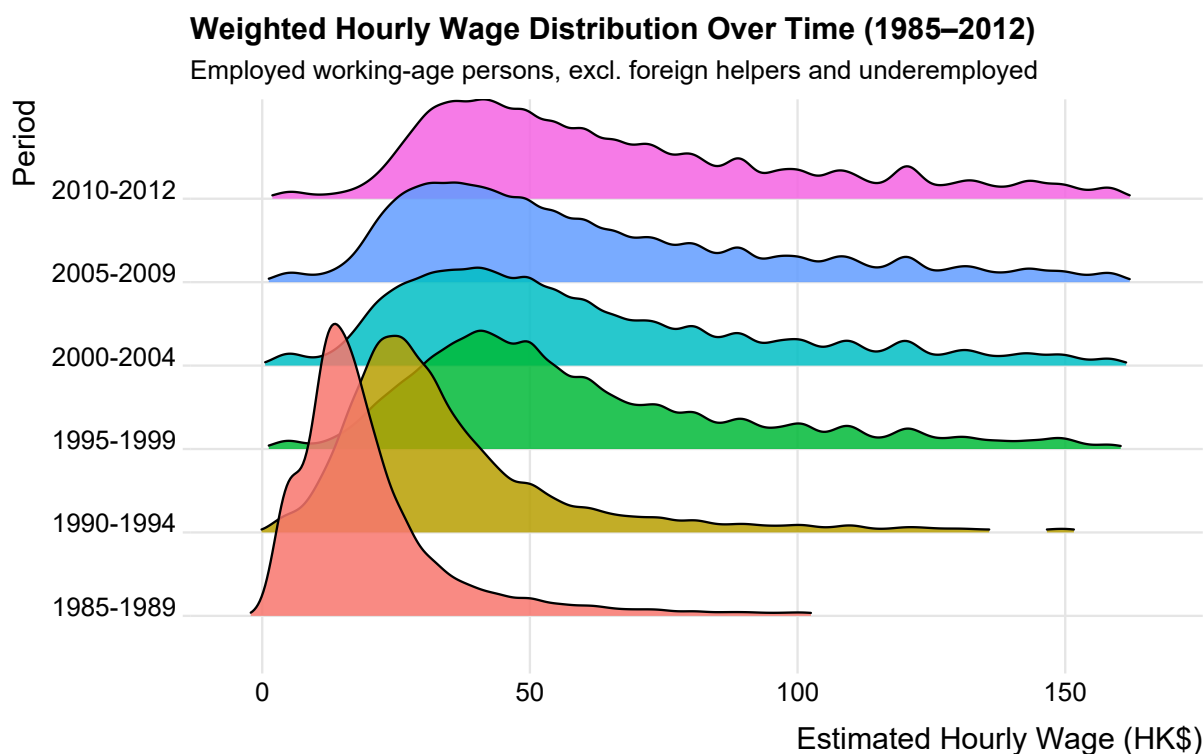
hours_7days * 4.345 is monthly working hour

care more about hours_main_employment first cuz earnings_main is for the main job, earnings_employed is total earnings

filter stuff

Nominal Wage

Figure 1: Long term wage growth



From 1985–1989 the distribution was heavily concentrated at very low wages (peak around HK\$20–30/hour). Over time, it shifted strongly to the right. By 2010–2012, the peak is around HK\$40–60/hour, and very few people are below HK\$30.

Hong Kong experienced strong long-term wage growth from 1985 to 2012. The hourly wage distribution shifted significantly to the right over the decades, with the lower tail becoming much thinner. By the late 2000s, very few workers were earning extremely low wages. When the Statutory Minimum Wage of HK\$28 per hour was introduced in May 2011, the wage floor was already relatively high for most employed working-age persons. As a result, we do not observe a dramatic visible shift in the overall wage distribution immediately around 2011. The policy likely had limited aggregate impact because it was not highly binding for the majority of the workforce.

Figure 2: Weighted Lower Tail Quantiles

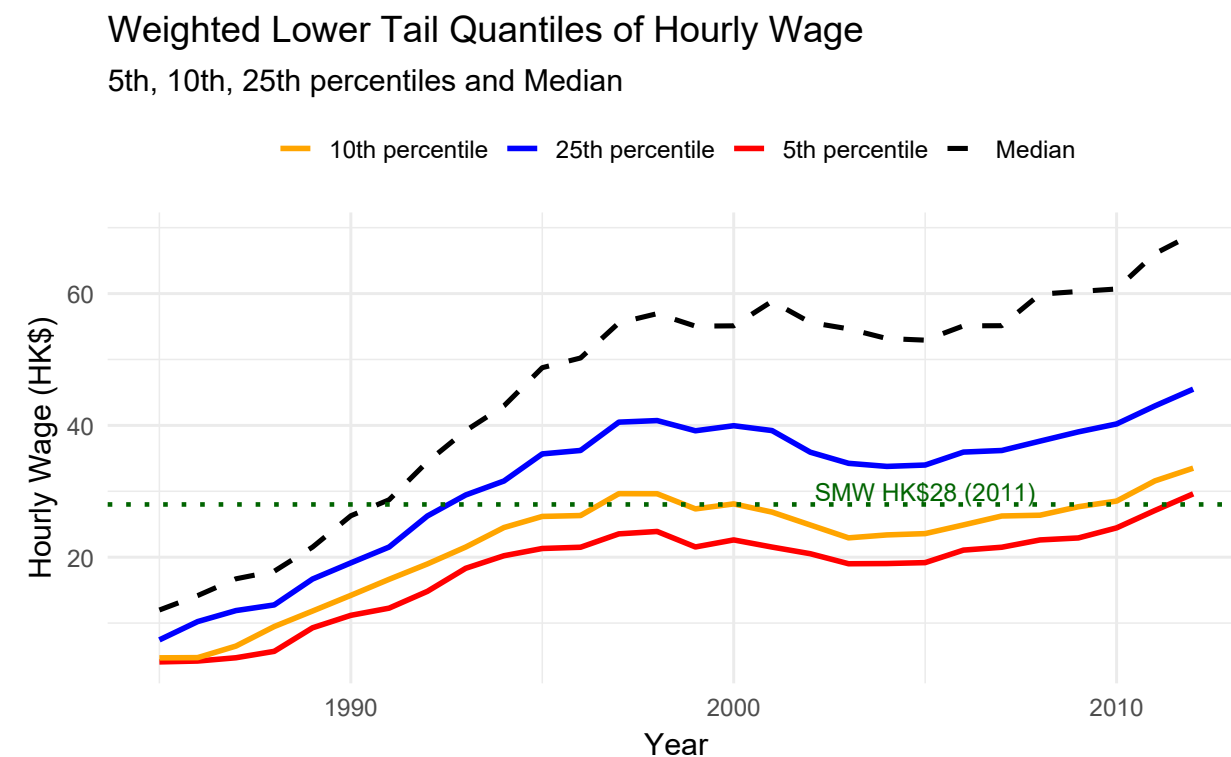


Figure 3: Weighted Proportion Below Thresholds

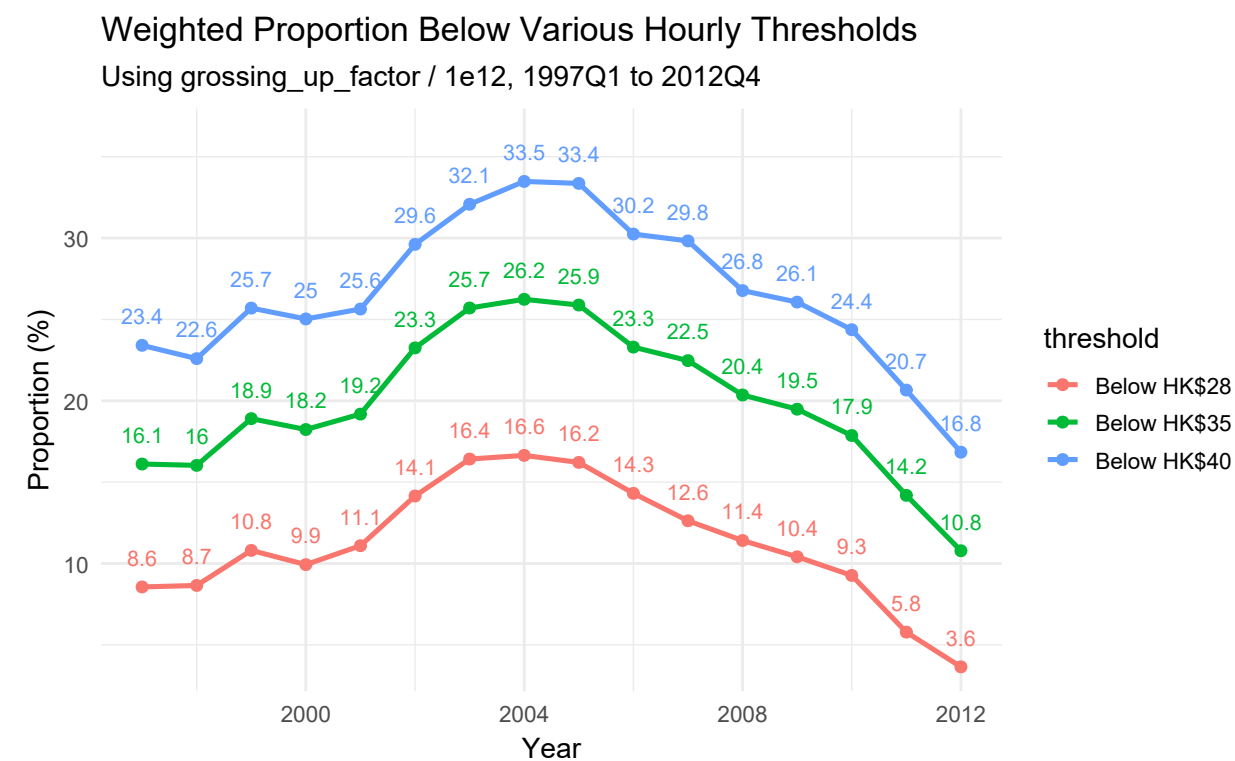


Fig3's YoY % change

year	Below HK\$28	Below HK\$35	Below HK\$40
2008	-9.61	-9.4	-10.2
2009	-8.77	-4.29	-2.64
2010	-11	-8.31	-6.5
2011	-37.6	-20.6	-15.2
2012	-36.9	-24	-18.5

if the % change is <0 i.e. decrease then red in color, else black, obviously the trend in above pic is always decreasing so only red exist

Below for Diagnostic Uses, Check 2011 numbers from both methods on the SAME df_focus

Can ignore

The reason this exist is i didnt use type = "i/n" in fig 2 so caused 2011 no one below min wages which is not same with fig 3's 2011

```
## Number of observations in 2011: 21242
```

```
## Sum of weights in 2011: 12.44582
```

```
## 5th percentile in 2011: 27.09
```

```
## Proportion below HK$28 in 2011: 5.78 %
```

```
## 5.8th percentile in 2011: 28.04
```

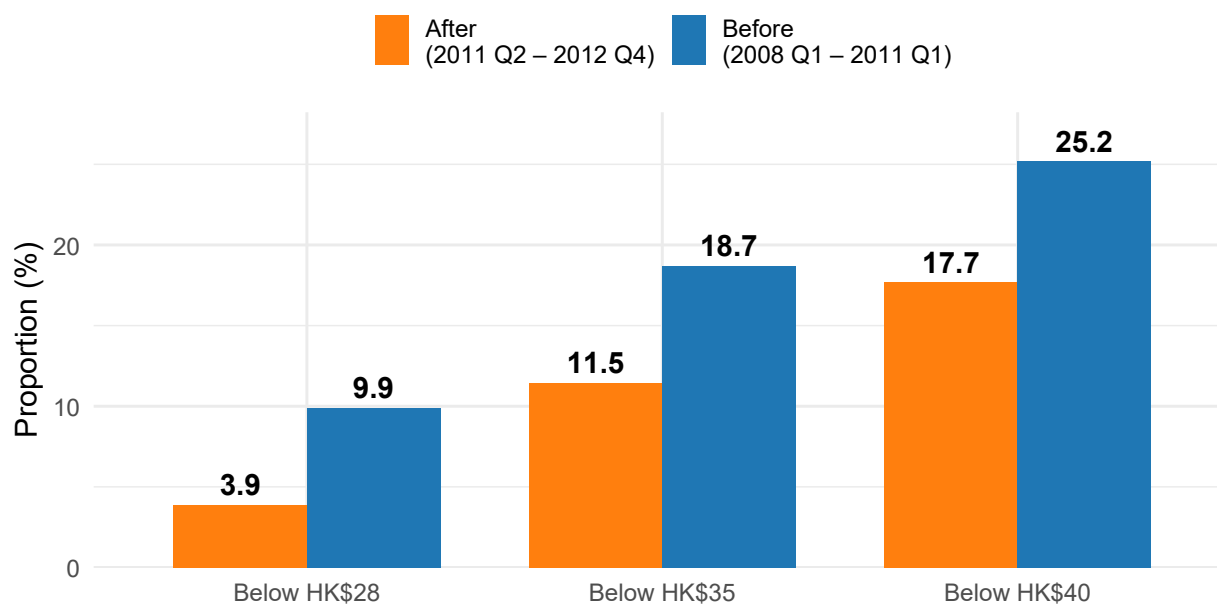
5th percentile is now HK\$27.09, which is logically consistent with 5.78% of the population being below HK\$28.

Figure 4: Weighted Before vs After

```
## # A tibble: 2 x 4
##   period                `Below HK$28` `Below HK$35` `Below HK$40`
##   <chr>                <dbl>      <dbl>      <dbl>
## 1 "After\n(2011 Q2 - 2012 Q4)"      3.89       11.5       17.7
## 2 "Before\n(2008 Q1 - 2011 Q1)"     9.92       18.7       25.2
```

Proportion Below Hourly Thresholds: Before vs After May 2011

Weighted | 2008 Q1 – 2011 Q1 vs 2011 Q2 – 2012 Q4



In the period before the minimum wage (2008 Q1 – 2011 Q1), approximately 9.92% of employed working-age persons earned below HK\$28 per hour. After the policy was implemented (2011 Q2 – 2012 Q4), this proportion dropped to 3.89% — a decline of 6.03 percentage points. Similar reductions are observed for the HK\$35 and HK\$40 thresholds

This suggests that after the Statutory Minimum Wage came into effect, fewer workers were earning very low hourly wages.

We now have:

- Long-term rightward shift in the wage distribution (Image 1)
- Clear improvement in the lower tail quantiles, with the 5th percentile reaching ~27.09 in 2011 (Image 2)
- A visible drop in the proportion below HK\$28 from ~9.3% in 2010 to 5.8% in 2011 (Image 3)
- A 6 percentage point drop in the before vs after comparison (Image 4)

There appears to be some acceleration in the compression of the lower tail around 2010–2012, coinciding with the introduction of the Statutory Minimum Wage in May 2011. The proportion of workers earning below HK\$28 dropped noticeably from 9.3% in 2010 to 5.8% in 2011, while the 5th percentile reached approximately HK\$27.09 in 2011.

Real Wage (CPI)

The graphs gen are nominal wage based

To get real wage we wil use cpi

We use https://www.censtatd.gov.hk/en/web_table.html?id=510-60001&full_series=1#

This gives us hk consumer price index MONTHLY

Year	Month	Comp_Index4	...	5	CPIA_Index7	...	8	...	9	...	10	...	11	...	12	...	13	...	14
1985	Jan	28.8	4.2	-0.1	28.7	3.3	-0.1	29.8	3.9	0.0	[3]	27.5	5.5	-0.4					
1985	Feb	28.9	3.5	0.5	28.9	2.8	0.8	30.0	3.4	0.6		27.7	4.7	0.6					
1985	Mar	29.2	4.4	0.7	29.2	4.1	0.9	30.2	4.4	0.7		27.8	4.8	0.3					
1985	Apr	29.3	4.1	0.6	29.3	3.9	0.4	30.4	4.2	0.6		28.0	4.3	0.9					
1985	May	29.2	3.5	-0.4	29.1	3.2	-0.6	30.3	3.6	-0.4		28.0	3.8	-0.1					
1985	Jun	29.4	3.1	0.5	29.3	2.8	0.5	30.4	3.2	0.4		28.2	3.3	0.5					

but monthly not too suit so we use quarterly,

basically jan + feb + mar /3 = q1 cpi etc

Year	quarter	cpi_comp	cpi_a
1985	1	28.96667	28.93333
1985	2	29.30000	29.23333
1985	3	29.50000	29.43333

[1] "2011 Q2 Reference CPI Value is: 78"

Real hourly wage = nominal hourly wage / current CPI * reference CPI

For years like 2012, where the CPI is higher than the 2011 reference (e.g., 82.5 vs 78.0), the calculation slightly reduces the salary.

The Logic: It shows that even if you earned more “paper money” in 2012, you couldn’t buy quite as much with it as you could in 2011 because prices went up.

For years like 1985, where the CPI is much lower than the reference (e.g., 28.9 vs 78.0), the calculation increases the nominal salary significantly.

The Logic: It reflects the fact that \$11 in 1985 had the same “buying power” as roughly \$30 in 2011. By boosting the 1985 number, we can see if that worker was actually “better off” than a minimum wage worker today.

The result: Even though Worker A earned fewer “dollars” (\$11 vs \$28), the math shows that Worker A was actually wealthier in real terms (\$29.68 vs \$28.00) because things were so much cheaper in 1985.

Example of Wage Adjustment (Real vs Nominal)

Table 1: Comparison of Nominal vs. Real Purchasing Power

Worker	Year	Nominal Wage	CPI	Calculation	Real Wage (2011 HKD)
Worker A	1985	11	28.9	(11.00 / 28.9) * 78.0	29.68
Worker B	2011	28	78.0	(28.00 / 78.0) * 78.0	28.00

Formula: Real hourly wage = nominal hourly wage / current CPI * reference CPI

Join the CPI quarterly data and calculate the Real Wage

Quick check to ensure the new column is there

year	quarter	hourly_wage	real_hourly_wage
2000	1	207.7739	227.08595
2000	1	249.3287	272.50314

year	quarter	hourly_wage	real_hourly_wage
2000	1	210.9705	230.57958
2000	1	105.8688	115.70903
2000	1	28.1294	30.74394
2000	1	560.9896	613.13207

Fig 1 Real wage long term growth

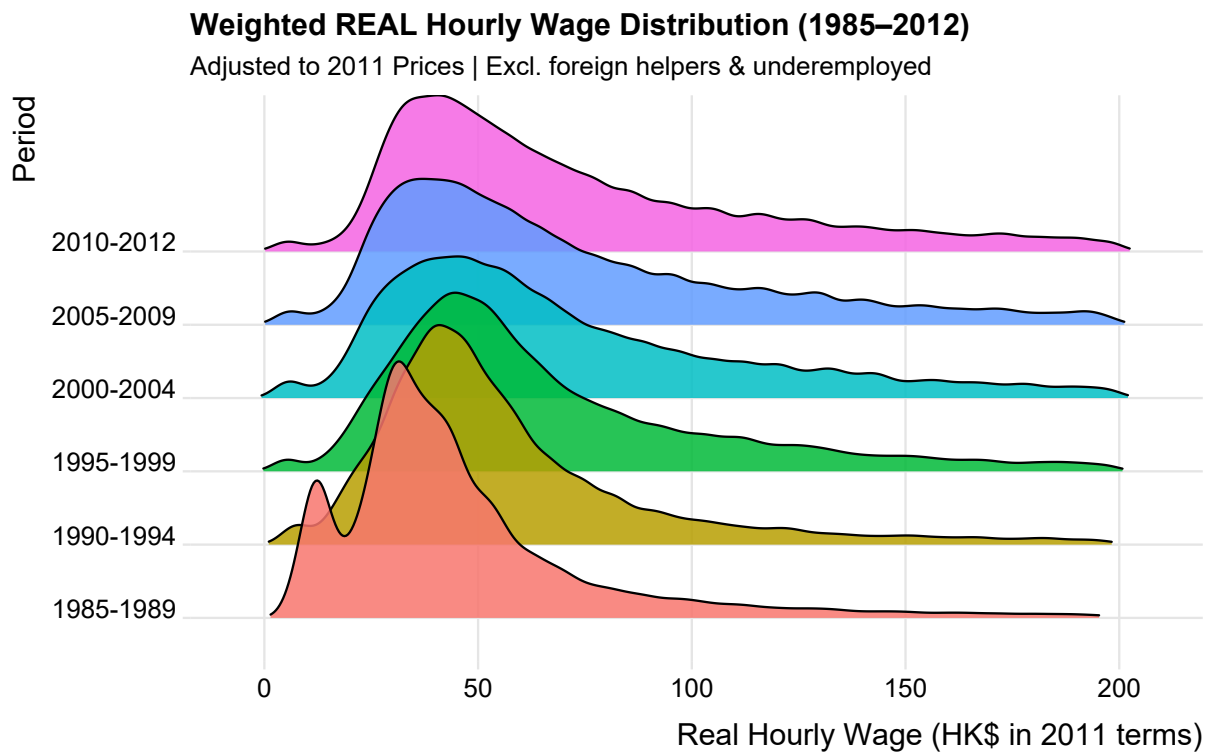


Figure 2: Real Wage Lower Tail Quantiles

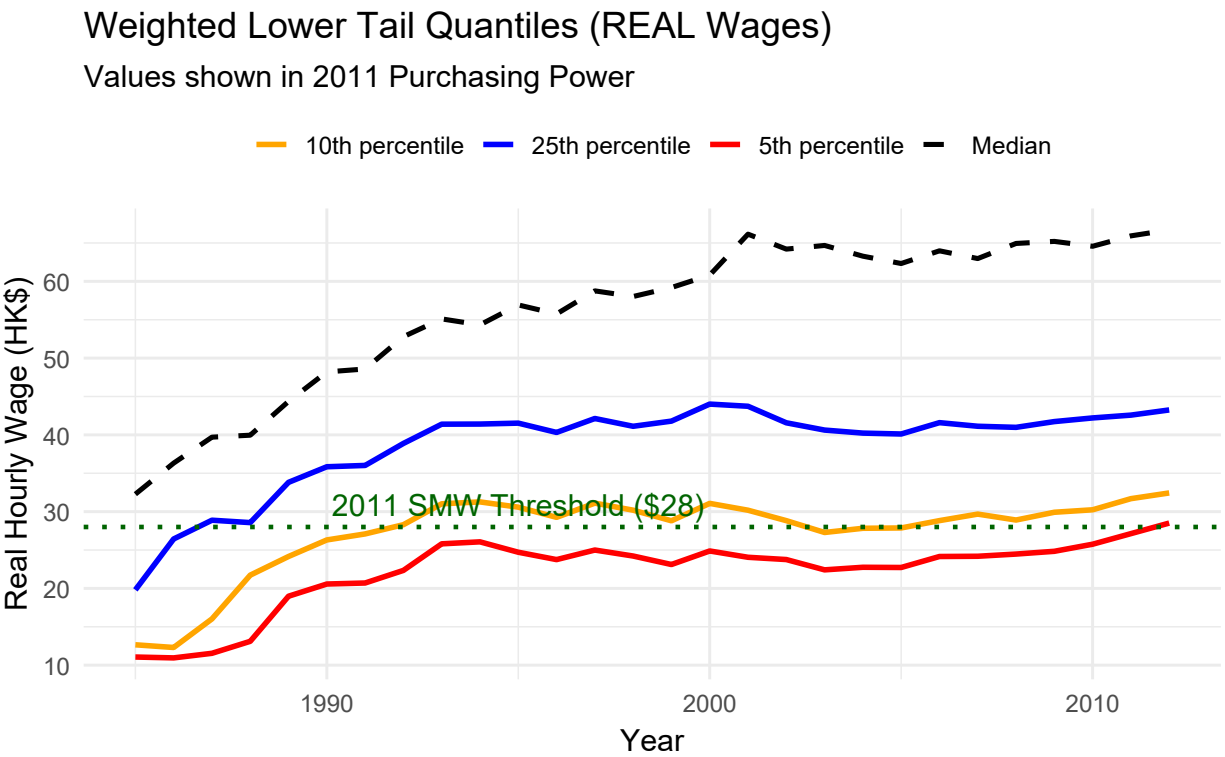


Figure 3: Proportion Below Real Thresholds

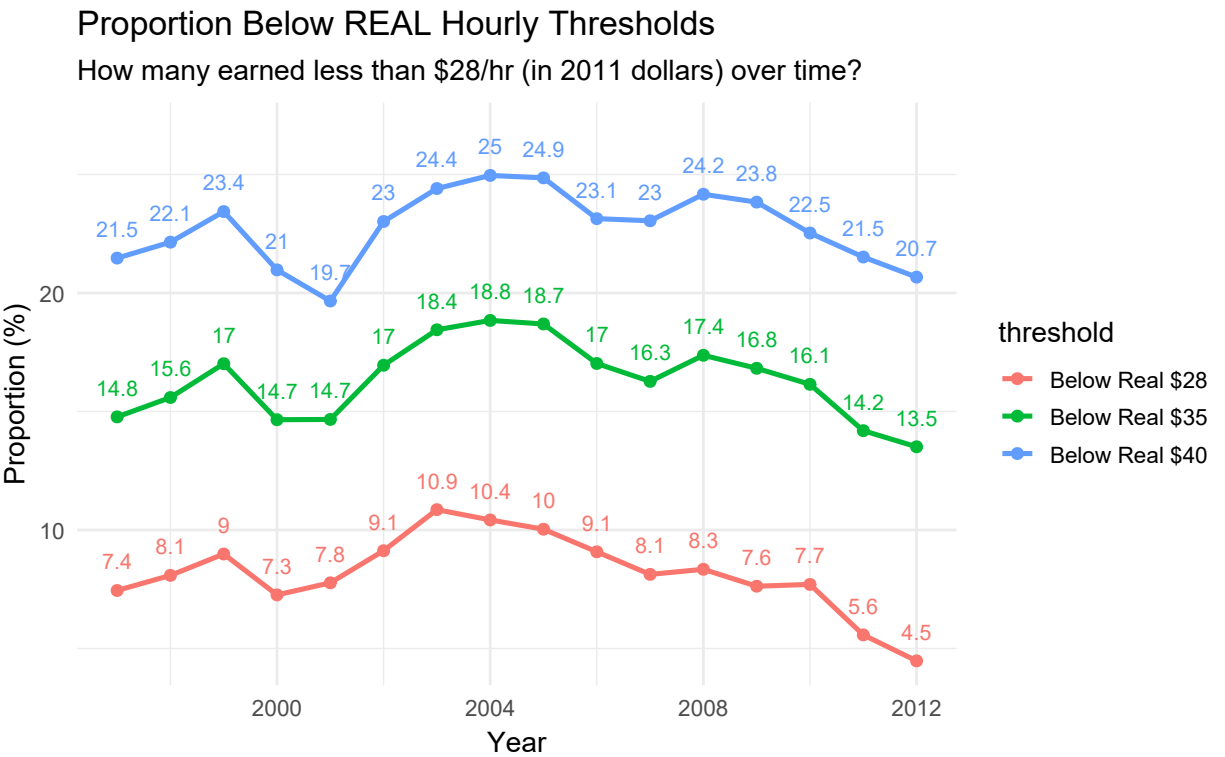
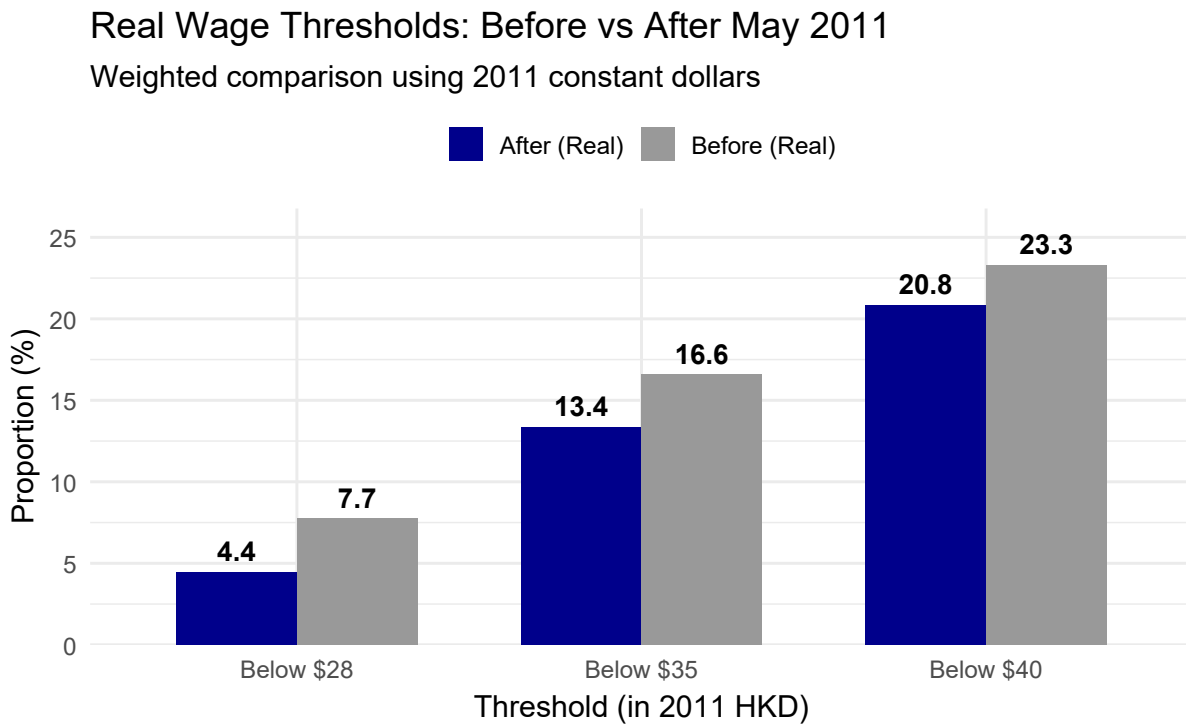


Fig3's YoY % change

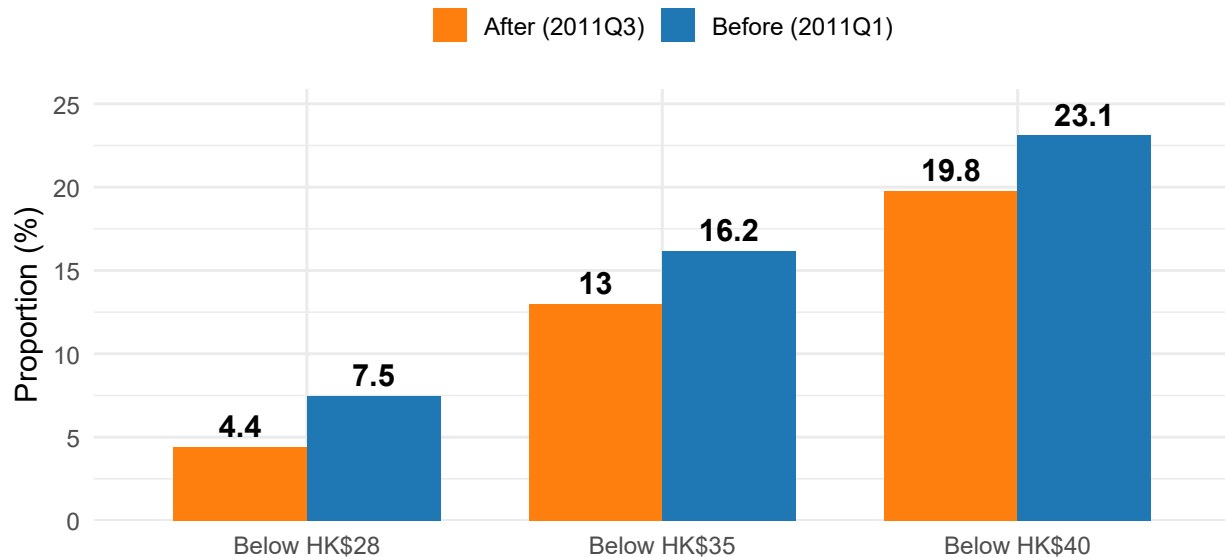
year	Below Real \$28	Below Real \$35	Below Real \$40
2008	2.65	6.77	4.86
2009	-8.56	-3.16	-1.38
2010	1	-4.03	-5.47
2011	-27.6	-12.1	-4.5
2012	-19.7	-4.8	-3.94

Figure 4: Before vs After (Real Wages)



```
## # A tibble: 2 x 4
##   period      `Below HK$28` `Below HK$35` `Below HK$40`
##   <chr>          <dbl>      <dbl>      <dbl>
## 1 After (2011Q3)      4.40        13.0        19.8
## 2 Before (2011Q1)     7.49        16.2        23.1
```

Proportion Below Hourly Thresholds: Right Before vs Right After May 2011 2011Q1 (Before) vs 2011Q3 (After) | All workers



Then further investigation

The aggregate data shows a continued improvement in wages and a reduction in the low-wage share around 2011, the effect appears modest in the overall population. This raises the question of whether the minimum wage had a stronger impact on specific vulnerable groups. Therefore, the next step is to conduct a more disaggregated analysis by examining subgroups such as:

- Workers with lower education levels
- Younger workers (e.g., aged 15–29) & prime workers maybe
- Employees in traditionally low-paying industries (e.g., retail, restaurants, cleaning) actually this idea is discarded cuz there could be some high pay people like owner of restaurants that also extremely high pay then will make it weird

By comparing the change in low-wage proportions before and after May 2011 across these groups, we can better understand the distributional effects of the policy and identify who benefited the most.

=== DATA QUALITY CHECKS ===

```
## === Basic Data Overview ===
## Total rows in raw data: 1300115
## Total rows after processing: 1300115
## Rows in focus sample: 572003
## Years covered:
## # A tibble: 112 x 3
##   year quarter     n
```

```

##      <dbl>      <dbl> <int>
## 1 1985          1 4502
## 2 1985          2 4807
## 3 1985          3 4698
## 4 1985          4 4769
## 5 1986          1 4732
## 6 1986          2 4832
## 7 1986          3 4659
## 8 1986          4 4683
## 9 1987          1 4627
## 10 1987         2 4743
## # i 102 more rows

##
## === Key Variable Summary ===

##   hourly_wage      real_hourly_wage      weight      hours_main_employment
## Min.   : 2.325   Min.   : 2.219   Min.   : 0.0003   Min.   : 8.00
## 1st Qu.: 27.170   1st Qu.: 37.459   1st Qu.: 0.0005   1st Qu.:40.00
## Median : 45.208   Median : 54.929   Median : 0.0005   Median :48.00
## Mean   : 69.905   Mean   : 83.719   Mean   :142.7027   Mean   :46.78
## 3rd Qu.: 79.114   3rd Qu.: 91.772   3rd Qu.: 0.0008   3rd Qu.:54.00
## Max.   :3452.244   Max.   :4090.254   Max.   :775.6073   Max.   :99.00
## earnings_main_hkd
## Min.   : 1000
## 1st Qu.: 5500
## Median : 9500
## Mean   : 13038
## 3rd Qu.: 15500
## Max.   :120000

##
## === Missing Values (%) ===

## # A tibble: 20 x 2
##   Variable      `Missing_%`
##   <chr>         <dbl>
## 1 industry_underemployed    100
## 2 previous_industry         100
## 3 previous_occupation        100
## 4 unemployment_duration      100
## 5 leave_reason               100
## 6 intend_same_industry       100
## 7 intend_industry            100
## 8 intend_same_occupation      100
## 9 intend_occupation          100
## 10 hours_secondary           99.7
## 11 year_completed_study      94.5
## 12 parent_serial             73.3
## 13 length_employment         58.8
## 14 spouse_serial             56.6
## 15 work_place                53.8
## 16 relationship              24.0
## 17 foreign_helper            24.0
## 18 other_employment_7days     24.0
## 19 education                  2.40

```

```
## 20 earnings_underemployed          1.84
##
## === Hourly Wage Distribution (Nominal) ===
```

Min	P5	Median	Mean	P95	Max	N
2.32	11.99	45.21	69.91	206.54	3452.24	572003

1. Check Earnings Mapping

```
## Unique earnings_main codes and their mapped values:
```

```
## # A tibble: 40 x 3
##   earnings_main earnings_main_hkd     n
##   <chr>          <dbl> <int>
## 1 01              1000 21222
## 2 02              2500 29452
## 3 03              3500 33208
## 4 04              4500 32707
## 5 05              5500 35875
## 6 06              6500 42288
## 7 07              7500 41219
## 8 08              8500 46906
## 9 09              9500 33480
## 10 10            10500 40181
## 11 11            11500 18784
## 12 12            12500 25518
## 13 13            13500 15513
## 14 14            14500  8794
## 15 15            15500 26255
## 16 16            16500  7180
## 17 17            17500  5605
## 18 18            18500  8614
## 19 19            19500  3158
## 20 20            21000 25134
## 21 21            23000  7187
## 22 22            25000 10608
## 23 23            27000  3379
## 24 24            29000  3022
## 25 25            31000 12607
## 26 26            33000  2149
## 27 27            35000  4402
## 28 28            37000  1147
## 29 29            39000  1011
## 30 30            41000  5255
## # i 10 more rows
```

2. Check Weighting Factor

```
## Total weighted population (should be around Hong Kong's employed population):
## [1] 81626372
```

3. Sample of Raw vs Processed Data

year	quarter	earnings_main	earnings_main_hk	hours_main_employment	hourly_wage	real_hourly_wage
1990	3	01	1000	36	6.39	11.70
1987	2	06	6500	55	27.20	66.37
2006	3	01	1000	12	19.18	22.05
2008	3	37	75000	45	383.58	414.78
1998	4	15	15500	54	66.06	68.25
2002	2	20	21000	36	134.25	154.30
1996	3	16	16500	39	97.37	107.22
1996	2	19	19500	44	102.00	113.33
1995	4	15	15500	51	69.95	80.19
1989	1	20	21000	41	117.88	250.08